

Maharashtra Public Service Commission Assessment

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Deputy Superintendent Assistant Commissioner of Police

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Deputy Superintendent Assistant Commissioner of Police

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Sub-Section Number :	1
Sub-Section Id :	4646627
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 1 Question Id : 464662401 Question Type : MCQ Option Shuffling : No

In the high frequency response of BJT amplifier, the upper cut-off (-3dB) frequency is mainly determined by _____.

Options :

1. Biasing resistors
2. Output capacitance
3. Power dissipation
4. Network capacitance and frequency dependence of $h_{fe}(B)$

Question Number : 2 Question Id : 464662402 Question Type : MCQ Option Shuffling : No

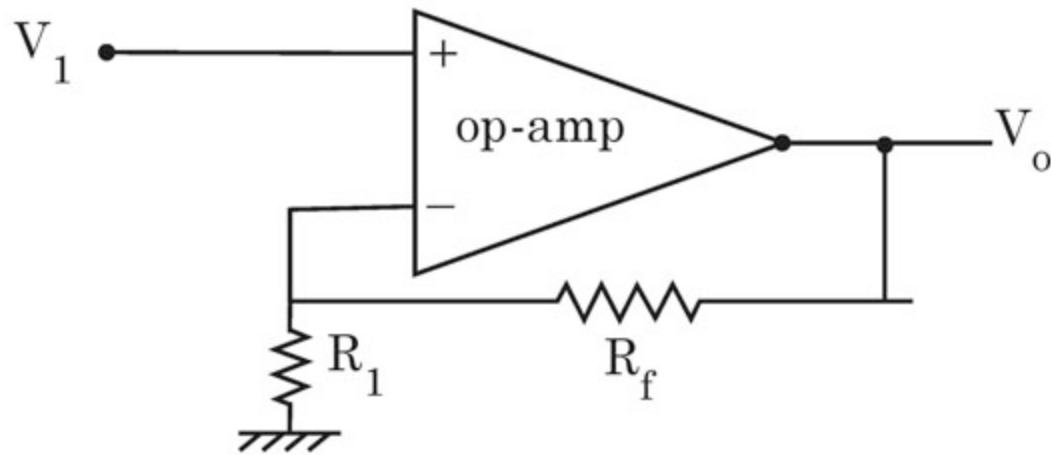
The 555 timer IC in a stable mode is used to generate _____.

Options :

1. A single fixed DC output
2. A continuous rectangular pulse waveform
3. A sine wave only
4. A DC regulated voltage

Question Number : 3 Question Id : 464662403 Question Type : MCQ Option Shuffling : No

What is the gain of the noninverting amplifier drawn below ?



Options :

1. $1 + \frac{R_f}{R_1}$
2. $-\frac{R_f}{R_1}$
3. $R_f \times R_1$
4. -1

Question Number : 4 Question Id : 464662404 Question Type : MCQ Option Shuffling : No

The condition in a JFET where the channel is fully constricted and drain current levels off is called _____.

Options :

1. Pinch-off
2. Breakdown
3. Saturation drift
4. Depletion

Question Number : 5 Question Id : 464662405 Question Type : MCQ Option Shuffling : No

One of the most important advantage in JK Master slave flip flop is _____.

Options :

1. It has preset
2. It has clock
3. It avoids racing
4. None of the above

Question Number : 6 Question Id : 464662406 Question Type : MCQ Option Shuffling : No

The logic family which has minimum power dissipation is _____.

Options :

1. TTL
2. I^2L
3. ECL
4. CMOS

Question Number : 7 Question Id : 464662407 Question Type : MCQ Option Shuffling : No

The degree of agreement within a group of measurements or instrument is referred as _____.

Options :

1. Accuracy
2. Resolution
3. Precision
4. Sensitivity

Question Number : 8 Question Id : 464662408 Question Type : MCQ Option Shuffling : No

Which one of the following transducer is self-generating transducer ?

Options :

1. Photomultiplier tube
2. Magnetic circuit transducer
3. Eddy current gage/gauge
4. Moving coil generator

Question Number : 9 Question Id : 464662409 Question Type : MCQ Option Shuffling : No

Electric flux density in Electric field is referred to as _____.

Options :

1. Number of flux lines
2. Ratio of flux lines crossing a surface and the surface area
3. Direction of flux at a point
4. Flux lines per unit area

Question Number : 10 Question Id : 464662410 Question Type : MCQ Option Shuffling : No

The force between two charges is 120 N. If the distance between the charges is doubled, the force will be _____.

Options :

1. 60 N
2. 30 N
3. 40 N
4. 15 N

Question Number : 11 Question Id : 464662411 Question Type : MCQ Option Shuffling : No

Which semiconductor material is used for mixed signal, DSP, RF, microwave, MM-wave applications ?

Options :

1. Silicon
2. Gallium Arsenide
3. Silicon Germanium
4. GaN

Question Number : 12 Question Id : 464662412 Question Type : MCQ Option Shuffling : No

Which is the part in radio system generally defined as everything between the antenna and the digital baseband system ?

Options :

1. RF front end
2. RF back end
3. IF front end
4. None of the above

Question Number : 13 Question Id : 464662413 Question Type : MCQ Option Shuffling : No

DFT is applied to _____.

Options :

1. Infinite sequences
2. Finite discrete sequences
3. Continuous infinite signals
4. Continuous finite sequences

Question Number : 14 Question Id : 464662414 Question Type : MCQ Option Shuffling : No

For each of the following systems, which system is a) Linear,
b) Time invariant and c) Causal ?

Options :

1. $y[n] = x[n] \cos(0.2\pi n)$

2. $y[n] = x[n] - x[n - 1]$

3. $y[n] = |x[n]|$

4. $y[n] = Ax[n] + B$, where A and B are nonzero constants

Question Number : 15 Question Id : 464662415 Question Type : MCQ Option Shuffling : No

What is the spectral density of white noise ?

Options :

1. varies with bandwidth
2. varies with frequency
3. constant
4. infinite

Question Number : 16 Question Id : 464662416 Question Type : MCQ Option Shuffling : No

The modulated carrier signal given in the passage is _____.

Options :

1. $s(t) = a(t) \sin (\omega_c t)$

2. $s(t) = a(t) \cos (\omega_c t + \phi)$

3. $s(t) = m(t) \cos (\omega_c t)$

4. $s(t) = A \cos (\omega_c t)$

Question Number : 17 Question Id : 464662417 Question Type : MCQ Option Shuffling : No

In frequency modulation, if carrier frequency is 815 KHz and maximum frequency is 815030 Hz and minimum frequency is 814970 Hz, then what would be frequency deviation ?

Options :

1. ± 30 KHz
2. ± 30 Hz
3. ± 60 Hz
4. ± 60 KHz

Question Number : 18 Question Id : 464662418 Question Type : MCQ Option Shuffling : No

In frequency modulation, if modulating signal is sine wave then which state of modulating signal causes highest frequency in modulated signal ?

Options :

1. Peak amplitude of positive half cycle of sine wave
2. Peak amplitude of negative half cycle of sine wave
3. Zero amplitude of sine wave
4. Half of peak amplitude of positive half cycle of sine wave

Question Number : 19 Question Id : 464662419 Question Type : MCQ Option Shuffling : No

In which multiple access scheme all users use full channel bandwidth ?

Options :

1. FDMA
2. TDMA
3. CDMA
4. None of the above

Question Number : 20 Question Id : 464662420 Question Type : MCQ Option Shuffling : No

Multi beam antennas are used to separate radio signals in _____.

Options :

1. CDMA
2. TDMA
3. FDMA
4. SDMA

Question Number : 21 Question Id : 464662421 Question Type : MCQ Option Shuffling : No

The antipodal signals are _____.

Options :

1. The signals that are 180° apart
2. The signals that are 90° apart
3. The signals that are 360° apart
4. The signals that are 45° apart

Question Number : 22 Question Id : 464662422 Question Type : MCQ Option Shuffling : No

The Intersymbol Interference (ISI) causes due to _____.

Options :

1. Overlapping of received pulses
2. Imperfect Filtering
3. System bandwidth constraints
4. All of the above

Question Number : 23 Question Id : 464662423 Question Type : MCQ Option Shuffling : No

In phase shift keying, the binary signal is used to switch the phase of carrier wave between

Options :

1. 0° and 90°
2. 0° and 360°
3. 0° and 180°
4. 0° and 45°

Question Number : 24 Question Id : 464662424 Question Type : MCQ Option Shuffling : No

The mutual information is zero if and only if _____.

Options :

1. The input and output symbols of the channel are statistically dependent
2. The input and output symbols of the channel are statistically independent
3. Both (1) and (2)
4. None of the above

Question Number : 25 Question Id : 464662425 Question Type : MCQ Option Shuffling : No

If some source symbols are likely to occur rarely, then _____.

Options :

1. Shorter Codewords can be assigned to them
2. Longer Codewords can be assigned to them
3. Shorter Codewords with parity bits can be assigned to them
4. Longer Codewords with parity bits can be assigned to them

Question Number : 26 Question Id : 464662426 Question Type : MCQ Option Shuffling : No

Time diversity is used in _____.

Options :

1. GSM mobile phone receiver
2. CDMA mobile phone receiver
3. GPS system
4. Satellite system

Question Number : 27 Question Id : 464662427 Question Type : MCQ Option Shuffling : No

Which type of fading channel is difficult to model ?

Options :

1. Frequency selective
2. Flat
3. Fast
4. Slow

Question Number : 28 Question Id : 464662428 Question Type : MCQ Option Shuffling : No

In 3G and future communication system UMTS is _____.

Options :

1. User Mobile Technology System
2. User Mobile Telecommunication System
3. Universal Mobile Telecommunications System
4. Universal Multiple Telecommunications System

Which of the following statements are true in case of microwaves ?

- a. Microwaves are unidirectional.
- b. Microwave propagation is line-of-sight.
- c. Very high frequency microwaves cannot penetrate walls.
- d. Microwaves having frequencies between 1 and 300 GHz.

Options :

- 1. Only a and b
- 2. Only b and c
- 3. Only a, b and c
- 4. a, b, c and d

Question Number : 30 Question Id : 464662430 Question Type : MCQ Option Shuffling : No

Which of the following are the 3 major elements considered in a link budget ?

Options :

1. Antenna size, Doppler shift, carrier wavelength
2. Transmitter noise, cloud cover, traffic load
3. Allocation of resources, signal loss sources and noise sources
4. User mobility, access delay and fading speed

Question Number : 31 Question Id : 464662431 Question Type : MCQ Option Shuffling : No

The major design goal of TCP/IP reference model is _____.

Options :

1. To define higher level data structures
2. To connect multiple networks in a seamless way
3. To compress the data
4. To define syntax and symantics of data

Question Number : 32 Question Id : 464662432 Question Type : MCQ Option Shuffling : No

Which type of propagation should be used in Amplitude Modulation (AM) broadcasting to transmit the signal till 150 km ?

Options :

1. Ground-wave propagation
2. Sky-wave propagation
3. Line-of-sight propagation
4. None of the above

Question Number : 33 Question Id : 464662433 Question Type : MCQ Option Shuffling : No

Walkie-talkies and CB radios are both _____.

Options :

1. Half duplex system
2. Simplex system
3. Full duplex system
4. Wired system

Question Number : 34 Question Id : 464662434 Question Type : MCQ Option Shuffling : No

In International Standards Organizations (ISO) OSI model, OSI refers to _____.

Options :

1. Open Standard International
2. Open Systems Interconnection
3. Open Source International
4. None of the above

Question Number : 35 Question Id : 464662435 Question Type : MCQ Option Shuffling : No

The main function of integral action in PID controller is to _____.

Options :

1. Reduce overshoot only
2. Increase damping
3. Eliminate steady state error
4. Reduce rise-time only

Question Number : 36 Question Id : 464662436 Question Type : MCQ Option Shuffling : No

In convolutional code, the encoder is described as _____.

Options :

1. Combinational circuit
2. Sequential circuit
3. Digital filter
4. Linear block code

Question Number : 37 Question Id : 464662437 Question Type : MCQ Option Shuffling : No

In practical sampling, sampling rate f_s is _____.

Options :

1. $f_s = 2f_m$

2. $f_s \leq 2f_m$

3. $f_s \geq 2f_m$

4. $f_s > 2f_m$

Question Number : 38 Question Id : 464662438 Question Type : MCQ Option Shuffling : No

Pulse amplitude modulation is _____.

Options :

1. Digital modulation scheme
2. Pulse digital modulation scheme
3. Pulse Analog modulation scheme
4. Pulse code modulation scheme

Question Number : 39 Question Id : 464662439 Question Type : MCQ Option Shuffling : No

If a random process is strictly stationary then _____.

Options :

1. All probability distributions are invariant to the shift in time
2. It's mean and autocorrelation are independent of time
3. It's power spectral density is constant
4. Expected value is zero

Question Number : 40 Question Id : 464662440 Question Type : MCQ Option Shuffling : No

(7, 4) Hamming code can _____.

Options :

1. Detect and correct one error
2. Detect two errors and correct one error
3. Detect and correct two errors
4. Detect only errors and has no correction ability

Question Number : 41 Question Id : 464662441 Question Type : MCQ Option Shuffling : No

Which material is most commonly used for long wavelength pin and avalanche photodiodes ?

Options :

1. Silicon
2. Germanium
3. Ga As
4. In Ga As

Question Number : 42 Question Id : 464662442 Question Type : MCQ Option Shuffling : No

For a geostationary satellite link, the free space loss (L_{Fs}) is typically of the order of _____.

Options :

1. 20 dB
2. 100 dB
3. 500 dB
4. 200 dB

Question Number : 43 Question Id : 464662443 Question Type : MCQ Option Shuffling : No

The multi-hop, infrastructure-less wireless networks are known as _____.

Options :

1. Mobile ad hoc networks (MANETS)
2. Wireless mesh networks
3. Cellular Networks
4. Wi-MAX

Question Number : 44 Question Id : 464662444 Question Type : MCQ Option Shuffling : No

The peak data rate expected in IMT-2020 is _____.

Options :

1. 20 Gbps
2. 100 Gbps
3. 1 Tbps
4. 10 Mbps

Question Number : 45 Question Id : 464662445 Question Type : MCQ Option Shuffling : No

Signal Variability in wireless communication characterizes _____.

Options :

1. Data rate variation
2. Channel Bandwidth
3. Fading nature of the channel
4. Antenna efficiency

Question Number : 46 Question Id : 464662446 Question Type : MCQ Option Shuffling : No

In wireless 3G Network, the controlling of several cell base transceiver stations is done by _____.

Options :

1. Radio Access Controller
2. Radio Network Controller
3. Mobile Nodes
4. Routers

Question Number : 47 Question Id : 464662447 Question Type : MCQ Option Shuffling : No

The circles in the Smith chart pass through the point _____.

Options :

1. $(0, 1)$
2. $(0, -1)$
3. $(-1, 0)$
4. $(1, 0)$

Question Number : 48 Question Id : 464662448 Question Type : MCQ Option Shuffling : No

What is the purpose of an antenna balun ?

Options :

1. To increase the overall directivity of an antenna
2. To match a balanced antenna to an unbalanced transmission line
3. To alter the polarization from circular to linear
4. To protect the receiver from lightning strikes

Question Number : 49 Question Id : 464662449 Question Type : MCQ Option Shuffling : No

Which protocol is used in conjunction with SIP for media streaming ?

Options :

1. HTTP
2. SNMP
3. FTP
4. RTC

Question Number : 50 Question Id : 464662450 Question Type : MCQ Option Shuffling : No

In VoIP signalling protocol used is _____.

Options :

1. SIP
2. UDP
3. TCP
4. SS7

Question Number : 51 Question Id : 464662451 Question Type : MCQ Option Shuffling : No

How many links are required for sin devices in a mesh topology network ?

Options :

1. 10
2. 12
3. 15
4. 18

Question Number : 52 Question Id : 464662452 Question Type : MCQ Option Shuffling : No

A special buffer storage, smaller and faster than main storage, that is used to hold a copy of instructions and data in main storage that are likely to be needed next by the processor and that have been obtained automatically from main storage _____.

Options :

1. EEROM
2. Cache Memory
3. DRAM
4. SRAM

Question Number : 53 Question Id : 464662453 Question Type : MCQ Option Shuffling : No

In computers the number of bits used to represent data type such as integers and characters is part of _____.

Options :

1. Computer Organization
2. Compiler Design
3. Computer Architecture
4. Operating System

Question Number : 54 Question Id : 464662454 Question Type : MCQ Option Shuffling : No

In Java, which block is always executed, however this is an optional clause ?

Options :

1. try
2. catch
3. throw
4. finally

Question Number : 55 Question Id : 464662455 Question Type : MCQ Option Shuffling : No

In C++ what happens if a variable is defined more than once in same block, what it will do ?

Options :

1. Run time error
2. Compile time error
3. It will allow all declarations
4. None of these

Question Number : 56 Question Id : 464662456 Question Type : MCQ Option Shuffling : No

What is the role of constructor in classes ?

Options :

1. To modify the data whenever required
2. To destroy an object
3. To initialize the data members of an object when it is created
4. To call private functions from the outer world

Question Number : 57 Question Id : 464662457 Question Type : MCQ Option Shuffling : No

Which of the following is a language for specifying the database schema and other additional properties of the data ?

Options :

1. Data – Definition Language
2. Data – Manipulation Language
3. Data – Derivation Language
4. Data – Query Language

Question Number : 58 Question Id : 464662458 Question Type : MCQ Option Shuffling : No

We say that a relation R is in _____, if the domains of all attributes of R are atomic.

Options :

1. Second Normal Form
2. Third Normal Form
3. First Normal Form
4. Fourth Normal Form

Question Number : 59 Question Id : 464662459 Question Type : MCQ Option Shuffling : No

A _____ is a collection of interrelated data and a set of programs to access those data.

Options :

1. Entity Relation Diagram (ERD)
2. Flow Chart
3. Attribute
4. Database Management System (DBMS)

Question Number : 60 Question Id : 464662460 Question Type : MCQ Option Shuffling : No

According to software engineering principles a good design should exhibit _____.

Options :

1. Speed, memory and security
2. Firmness, commodity and delight
3. Cost, quality and reliability
4. Modularity, testing and maintenance

Question Number : 61 Question Id : 464662461 Question Type : MCQ Option Shuffling : No

What is cohesion in software design ?

Options :

1. Cohesion is a measure of interface complexity between modules
2. Cohesion is a measure of interconnection among modules
3. Cohesion is a measure of functional strength of a module
4. Cohesion is a measure of relative interdependence among modules

Question Number : 62 Question Id : 464662462 Question Type : MCQ Option Shuffling : No

During unit testing, which testing activity should be performed before any other ?

Options :

1. Boundary condition testing
2. Independent path testing
3. Error-handling path testing
4. Testing of data flow across the component interface

Question Number : 63 Question Id : 464662463 Question Type : MCQ Option Shuffling : No

Which graph algorithm detects negative weight cycle ?

Options :

1. Dijkstra
2. Prim
3. Kruskal
4. Bellman-ford

Question Number : 64 Question Id : 464662464 Question Type : MCQ Option Shuffling : No

While hashing collision resolution technique that uses linked lists is _____.

Options :

1. Open addressing
2. Double hashing
3. Linear probing
4. Chaining

Question Number : 65 Question Id : 464662465 Question Type : MCQ Option Shuffling : No

An algorithm searches for a key in a sorted array by repeatedly dividing the search space into half. What will be the best case time complexity ?

Options :

1. $O(n)$
2. $O(n^2)$
3. $O(\log n)$
4. $O(1)$

Question Number : 66 Question Id : 464662466 Question Type : MCQ Option Shuffling : No

To find the MAC address of the next hop, IP needs which protocol ?

Options :

1. ICMP
2. RARP
3. ARP
4. DHCP

Question Number : 67 Question Id : 464662467 Question Type : MCQ Option Shuffling : No

Domain Name System (DNS) query and response messages between a resolver and a local DNS server are transmitted using _____.

Options :

1. UDP packets
2. TCP segments
3. ICMP messages
4. SCTP association

Question Number : 68 Question Id : 464662468 Question Type : MCQ Option Shuffling : No

A company employee sends an official email with attachments from outlook to a colleague in another country. Which protocol is used to send this email ?

Options :

1. POP3
2. SMTP
3. IMAP
4. FTP

Question Number : 69 Question Id : 464662469 Question Type : MCQ Option Shuffling : No

Which organization is responsible for allocating large blocks of IP addresses at the highest level ?

Options :

1. ISP
2. DNS
3. ICANN
4. IEEE

Question Number : 70 Question Id : 464662470 Question Type : MCQ Option Shuffling : No

Which of the following Wi-Fi networks commonly used in universities and classrooms ?

Options :

1. Ad hoc networks
2. Mesh networks
3. Single hop infrastructure-based networks
4. Bluetooth networks

Question Number : 71 Question Id : 464662471 Question Type : MCQ Option Shuffling : No

In a messaging app, users sign messages, so recipients can confirm the sender's identity without sharing secret keys. Which system enables this process ?

Options :

1. Symmetric key signature system
2. Hash-based verification
3. Public-key digital signature system
4. Physical authentication

Assertion (A) : The SSL Record Protocol provides two services for SSL connections : confidentiality and message integrity.

Reason (R) : The Handshake Protocol defines a shared secret key that is used for conventional encryption of SSL payloads.

Which of the following statement is correct ?

Options :

1. Both (A) and (R) are true and (R) is the correct explanation of (A)
2. Both (A) and (R) are true but (R) is not the correct explanation of (A)
3. (A) is true but (R) is false
4. (A) is false but (R) is true

Question Number : 73 Question Id : 464662473 Question Type : MCQ Option Shuffling : No

In PGP, there can be how many paths from fully or partially trusted authorities to any subject ?

Options :

1. Single path
2. Multiple paths
3. Single and multiple paths
4. No path

Question Number : 74 Question Id : 464662474 Question Type : MCQ Option Shuffling : No

In which penetration test, the testing team know, no details of the inner working of the application ?

Options :

1. White box testing
2. Black box testing
3. Gray box testing
4. Blue box testing

Question Number : 75 Question Id : 464662475 Question Type : MCQ Option Shuffling : No

What is the value of result after executing the following Java Script code

let result = 3 + 4 * 2/8

 ?

Options :

1. 3.5
2. 4
3. 4.5
4. 5

Question Number : 76 Question Id : 464662476 Question Type : MCQ Option Shuffling : No

To prevent SQL Injection, a web application should primarily _____.

Options :

1. Use parameterized (bind) queries when executing SQL statements
2. Store SQL queries in separate files
3. Use the POST method instead of GET
4. Increase data base privileges

Question Number : 77 Question Id : 464662477 Question Type : MCQ Option Shuffling : No

Which statement correctly differentiates `require()` and `include()` constructs in PHP in case of failure ?

Options :

1. both `require()` and `include()` generates fatal errors on failure
2. `require()` generates a warning while `include()` generates a fatal error
3. both `require()` and `include()` ignore errors and continue execution
4. `require()` generates a fatal error, whereas `include()` generates a warning

Question Number : 78 Question Id : 464662478 Question Type : MCQ Option Shuffling : No

When you are accessing Google docs for accessing the features of a document writing software from your browser without specifically installing them. Which among the following cloud service models is the most appropriate one that you are using ?

Options :

1. SaaS
2. PaaS
3. IaaS
4. DaaS

Consider the following statements regarding the architectural difference between operating system level virtualization and hardware virtualization.

- S1 : Operating system level virtualization requires modified guest operating systems.
- S2 : Operating system level virtualization has no virtual machine manager or hypervisor and virtualization is implemented within a single operating system kernel.
- S3 : Operating system level virtualization works only on windows based systems.
- S4 : Operating system level virtualization depends on hardware assisted virtualization extensions.

Which of the above statements is/are correct ?

Options :

1. S1 and S2 only
2. S2 only

3. S2 and S4 only

4. S1, S2 and S4 only

Match the following related to amazon web services and their use.

- | | |
|-----------------------|---|
| a. Amazon Cloud Watch | I. Highly durable and distributed data store |
| b. Amazon S3 | II. Extend your corporate network into a private cloud inside AWS |
| c. Amazon Simple DB | III. Enable monitoring on an Amazon EC2 instance |
| d. Amazon VPC | IV. Provides the core functionality of a database |

Options :

- | | a | b | c | d |
|----|----------|----------|----------|----------|
| 1. | III | I | IV | II |

- | | a | b | c | d |
|----|----------|----------|----------|----------|
| 2. | I | III | II | IV |

- | | a | b | c | d |
|----|----------|----------|----------|----------|
| 3. | IV | III | II | I |

4. **a** **b** **c** **d**
 II III IV I

Question Number : 81 Question Id : 464662481 Question Type : MCQ Option Shuffling : No

What is a Blockchain ?

Options :

1. A centralized database
2. A networking protocol
3. A decentralized ledger
4. A type of database

Question Number : 82 Question Id : 464662482 Question Type : MCQ Option Shuffling : No

What is the fundamental goal of Real Time Transport Protocol (RTP) ?

Options :

1. to allow multimedia applications to work together
2. to give choice of audio and video compression techniques
3. in real time multimedia applications to absorb the time delays with retransmission
4. all of the above

Question Number : 83 Question Id : 464662483 Question Type : MCQ Option Shuffling : No

Two sub-layers of Constrained Application Protocol (CoAP) are _____.

Options :

1. Messaging and Holding
2. Messaging and Backoff
3. Messaging and Teardown
4. Messaging and Request/Response

Question Number : 84 Question Id : 464662484 Question Type : MCQ Option Shuffling : No

Which of the following is not a unsupervised machine learning algorithm ?

Options :

1. Linear Regression
2. K-means
3. Self Organising Map (SOM)
4. Apriori Algorithm

_____ is a machine learning approach that lets users play an important role.

Options :

1. Semi supervised learning
2. Active learning
3. Supervised learning
4. Unsupervised learning

Question Number : 86 Question Id : 464662486 Question Type : MCQ Option Shuffling : No

MIMO uses multiple antennas at the _____.

Options :

1. Transmitter only
2. Receiver only
3. Both transmitter and receiver
4. Relay only

Question Number : 87 Question Id : 464662487 Question Type : MCQ Option Shuffling : No

Under poor channel conditions Adaptive Modulation and Coding (AMC) generally selects _____.

Options :

1. Higher data rate
2. Lower data rate
3. No coding
4. Analog transmission

Question Number : 88 Question Id : 464662488 Question Type : MCQ Option Shuffling : No

Interrupt enable and disable mechanisms in RTOS are used primarily to protect _____.

Options :

1. Device registers
2. User-mode critical sections
3. Kernel memory
4. Scheduler data structures

Question Number : 89 Question Id : 464662489 Question Type : MCQ Option Shuffling : No

The primary goal of a cruise-control system is _____.

Options :

1. Increase fuel consumption
2. Maintain the vehicle near drivers desired speed
3. Prevent braking completely
4. Control the steering of the wheel

Question Number : 90 Question Id : 464662490 Question Type : MCQ Option Shuffling : No

A 16×8 CAM is to be designed using 8×2 CAM chips. How many chips, banks and address lines are required respectively _____.

Options :

1. 4 chips, 2 banks, 4 address lines
2. 8 chips, 2 banks, 4 address lines
3. 8 chips, 4 banks, 3 address lines
4. 16 chips, 2 banks, 5 address lines

Question Number : 91 Question Id : 464662491 Question Type : MCQ Option Shuffling : No

Consider the Newton Raphson method to find the root of the equation

$$f(x) = x^3 + 3x - 7 = 0, \text{ starting from } x_0 = 1.$$

What is the value of x_1 after the first iteration ?

Options :

1. 1.4
2. 1.6
3. 1.5
4. 1.75

Question Number : 92 Question Id : 464662492 Question Type : MCQ Option Shuffling : No

A complex number of the form $a + jb$ is called _____.

Options :

1. Polar Complex Number
2. Cartesian Complex Number
3. Exponential Complex Number
4. Trigonometric Complex Number

Question Number : 93 Question Id : 464662493 Question Type : MCQ Option Shuffling : No

The primary goal of Quality Function Deployment (QFD) is to _____.

Options :

1. Reduce production costs
2. Increase marketing campaigns
3. Translate customer needs into design requirements
4. Improve employee satisfaction

Question Number : 94 Question Id : 464662494 Question Type : MCQ Option Shuffling : No

Telecommunication sector developments such as 5G mainly impact on engineering applications by enabling _____.

Options :

1. Higher compiler efficiency
2. Low latency, reliability and high speed communication
3. Reduced power generation
4. Improved mechanical strength

Question Number : 95 Question Id : 464662495 Question Type : MCQ Option Shuffling : No

Which global sporting event did India host and win in late 2025 ?

Options :

1. Asia Cup T20
2. ICC Women's Cricket World Cup
3. Olympics 2025
4. Commonwealth Games

Question Number : 96 Question Id : 464662496 Question Type : MCQ Option Shuffling : No

A serious terrorist threat has enlarged and intelligence agencies need to intercept certain communication. This is permissible under the Indian Telegraphy Act if _____.

Options :

1. The telecom company agrees
2. There is a public emergency or public safety concerns and reasons are recorded in writing
3. Parliament passes a new law
4. Every citizen is informed

Question Number : 97 Question Id : 464662497 Question Type : MCQ Option Shuffling : No

Under the Indian Wireless Telegraphy Act, 1933 the Central Government has the power to _____.

Options :

1. Exempt any person/class from the Act only through separate notification, not rules
2. Determine by rules whether any article or class shall be treated as wireless telegraphy apparatus
3. Issue licenses directly, bypassing the telegraph authority constituted under the 1885 Act
4. Impose imprisonment for possession of any wireless apparatus (other than transmitter)

Question Number : 98 Question Id : 464662498 Question Type : MCQ Option Shuffling : No

Unauthorised disclosure of intercepted messages under the Telegraphy Act, 1885 is punishable with _____.

Options :

1. Fine only
2. Imprisonment upto 1 year
3. Imprisonment upto 3 years
4. Imprisonment upto 7 years

Question Number : 99 Question Id : 464662499 Question Type : MCQ Option Shuffling : No

Under the Indian Telegraphy Act, a person without an Aadhaar Number _____.

Options :

1. Cannot obtain telecom services
2. Cannot be denied service solely for not having Aadhaar
3. Must enroll for Aadhaar
4. Can obtain services only with court permission

Question Number : 100 Question Id : 464662500 Question Type : MCQ Option Shuffling : No

Under the Indian Telegraphy Act, 1885 the Central Government has the exclusive privilege to _____.

Options :

1. Manufacture telephones only
2. Establish, maintain and work telegraphs
3. Sell communication devices
4. Regulate internet content only